SmartGraphics Subpage Reference List Users Guide

1 SUMMARY

The Subpage Reference List is a SmartGraphics control that allows a designer to create a scrollable list, but reference an actual subpage as the list item.

The following minimum software required is:

- VTPRO-e Version 5.7 and greater.
- SmartGraphics Version 2.05.xx.xx and greater.

A training video has been created to assist with the design and integration of this control. Please visit our Vimeo channel to view the video: http://vimeo.com/crestron

2 DESIGN CONSIDERATIONS

The following should be considered, while designing subpages to be used with this control:

- Only one subpage can be referenced in the list.

  If the goal is to provide different object layouts within the subpage reference, take advantage of the visibility join property of each object. This will allow many different types of objects to be placed in the subpage, but made visible when needed/required.

- The list will adjust itself to the size of the subpage reference, based on the list orientation.

  A horizontal list will adjust its height to the height of the subpage reference. A vertical list will adjust its width to the width of the subpage reference.

- Only perpendicular gestured objects can be used in the subpage reference.

  It is extremely difficult to gesture an object that needs to be “swiped” in the same direction as the list movement. If the list orientation is vertical, a vertical object like Fader Slider Vertical, cannot be added to a subpage that will be referenced by the list. In this case, only horizontally oriented objects, like Fader Slider Horizontal, can be added.

- The join number gap for each type of join (digital, analog & serial) will be based on the highest join number of each object type.

This is one of the most important topics to understand, when using this object. If a button on the subpage reference is given a join of 500 and there are no other digital joins on the subpage, the join gap between each list item will then need to be 500. This will essentially waste 499 digital joins for each list item. For this reason, each join type should always start at 1.
See the Join Number Management section for more detail.

- Leave room for expansion

  When setting the join number gaps for the digital, analog and serial objects in the subpage reference, make sure extra joins are added to allow for the future addition of more objects on the subpage.

  See the Join Number Management section for more detail.

3 General Notes

There has been a lot of work to resolve issues of on-screen refresh of the list, but there may be times while the list is moving, that it may appear for a moment that a list item is missing. When the list stops moving, an update should occur that reflects the current state of all list items.

4 Join Number Management

As mentioned previously, understanding how joins are mapped and indexed is one of the most important topics to understand with this control.

Each join type (digital, analog and serial) has an offset, which is called "Join Increment". This value, set in the property grid at design time, determines the join number gap between the list items. The minimum value has to be set to the highest join number of a given type.

I.E. If the subpage reference has the following objects/joins

Type	Join Type	Join Value
Advanced Button	Digital	1
Advanced Button	Digital	2
Advanced Button	Digital	3
Advanced Button	Digital	4
Segmented Gauge	Analog	1
Segmented Gauge	Analog	2
Liquid Gauge	Analog	3
Formatted Text	Serial	1
Formatted Text	Serial	2

The minimum Join Increments should be set as follows:

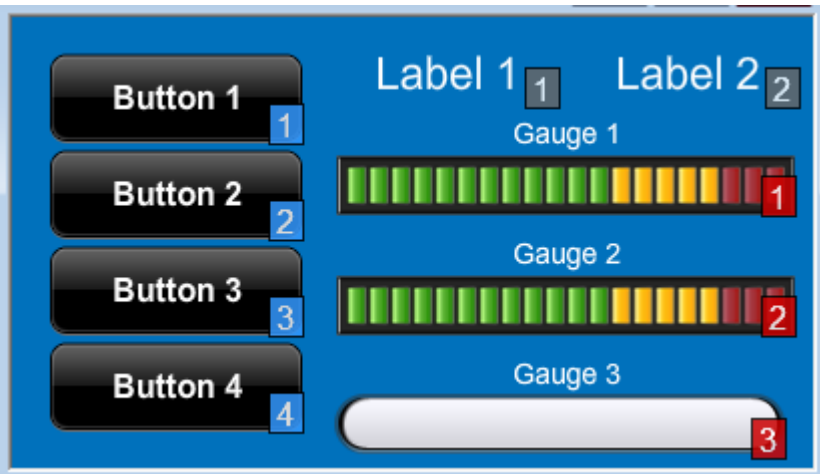
Digital Join Increment: 4

Analog Join Increment: 3
Serial Join Increment: 2

As stated previously however, those values should be “padded” to allow for future expansion.

5 Join Control In SIMPL Windows

Based on the example given in the Join Number Management section, here is the subpage:

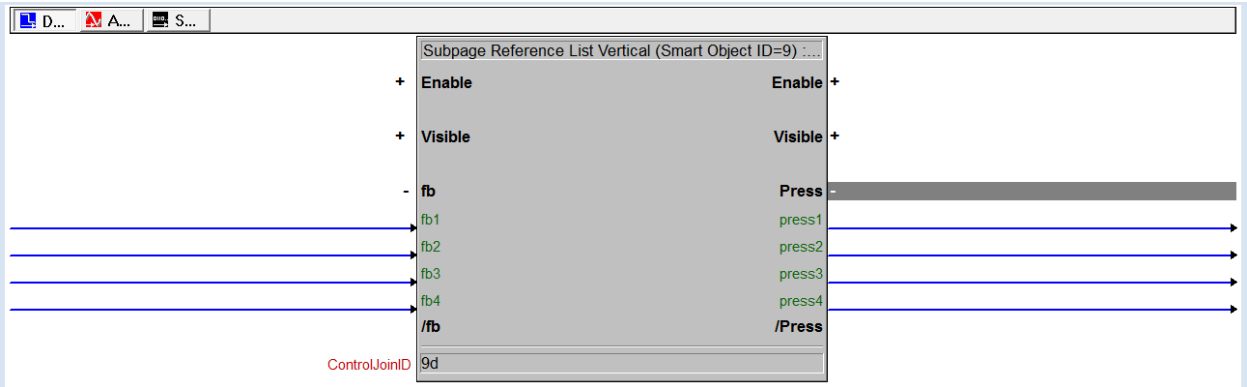


The join increments do not have any padding and are set as follows:

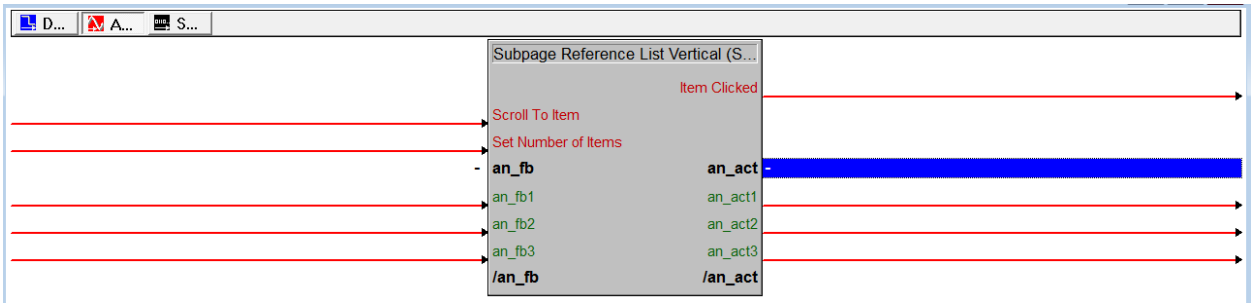
Digital Join Increment: 4
Analog Join Increment: 3
Serial Join Increment: 2

Setting the Number of Subpage References = 1 and then compiling the VTPRO project, creates the following SGD for SIMPL Windows:

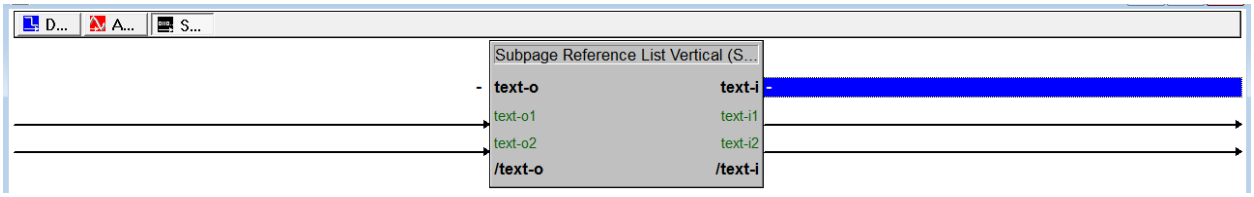
Digitals



Analogs



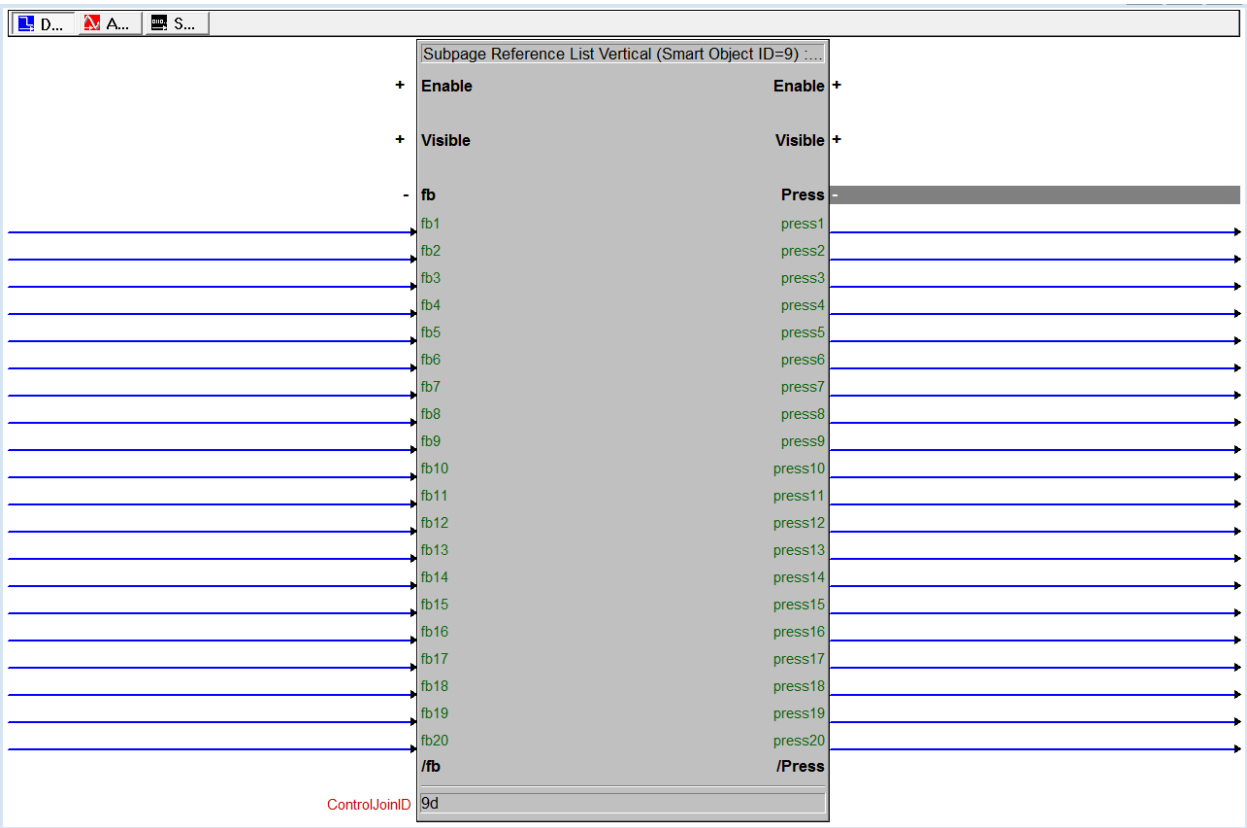
Serials



With only one (1) list item and no padding of the join increments, it is a 1:1 relationship of the digital, analog and serial joins to the actual objects on the subpage.

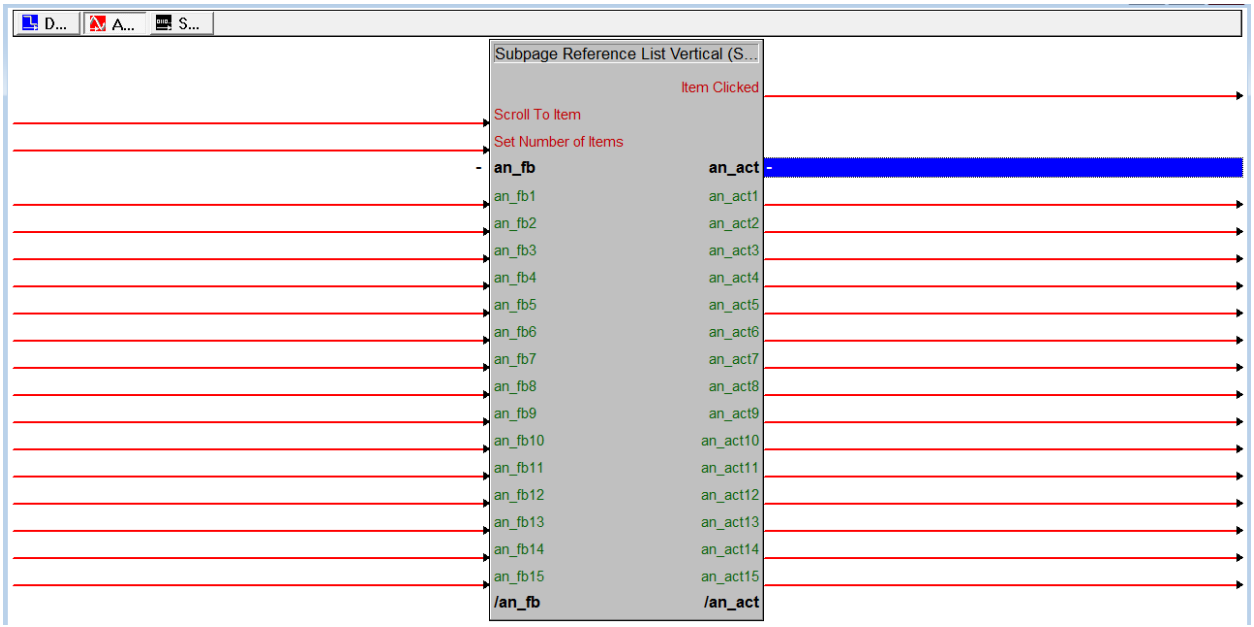
Increasing the Number of Subpage References = 5 and then compiling the VTPRO project, creates the following SGD for SIMPL Windows:

Digitals



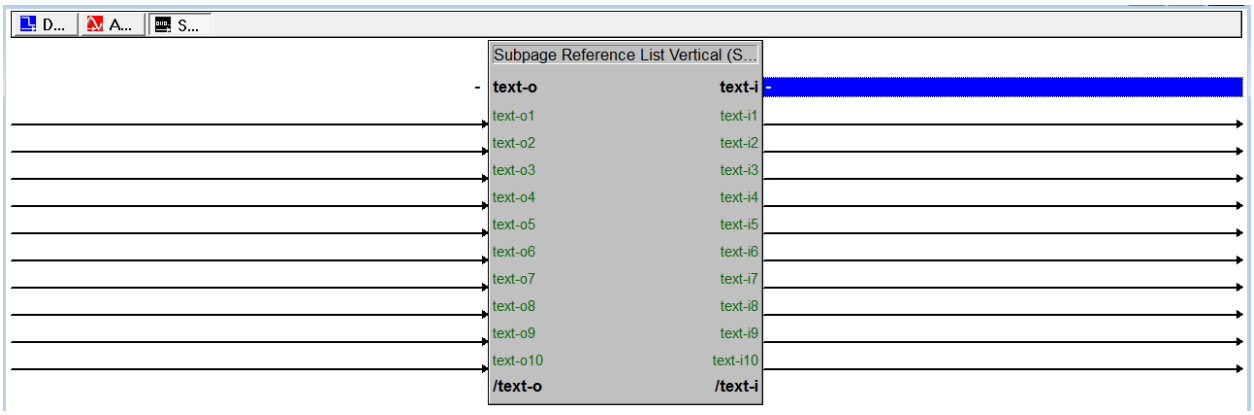
- press/fb 1 – 4 map to Buttons 1 – 4 on the first list item.
- press/fb 5 – 8 map to Buttons 1 – 4 on the second list item.
- press/fb 9 – 12 map to Buttons 1 – 4 on the third list item.
- Etc.

Analogs



- an_act/fb 1 – 3 maps to Gauges 1 – 3 on the first list item.
- an_act/fb 4 – 6 maps to Gauges 1 – 3 on the second list item.
- an_act/fb 7 – 9 maps to Gauges 1 – 3 on the third list item.
- Etc.

Serials



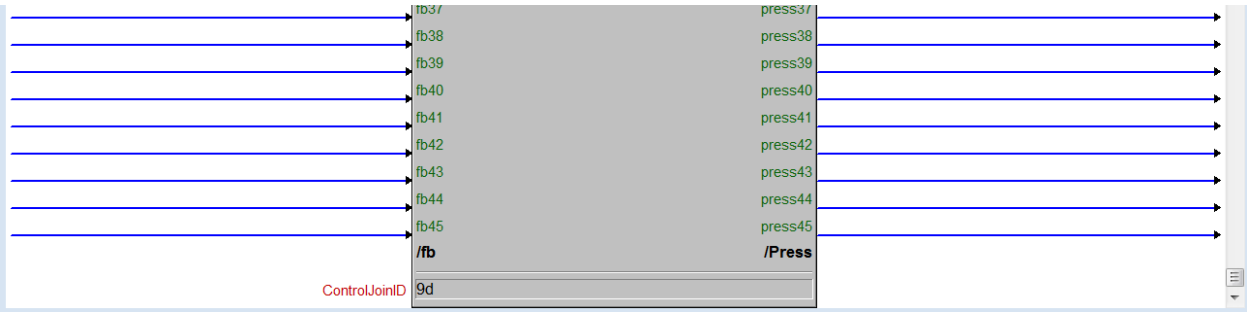
- Text-i/o 1 – 2 maps to Labels 1 – 2 on the first list item.
- Text-i/o 3 – 4 maps to Labels 1 – 2 on the second list item.
- Text-i/o 5 – 6 maps to Labels 1 – 2 on the third list item.
- Etc.

Padding each join increment by 5 join numbers:

Digital Join Increment: 9
Analog Join Increment: 8
Serial Join Increment: 7

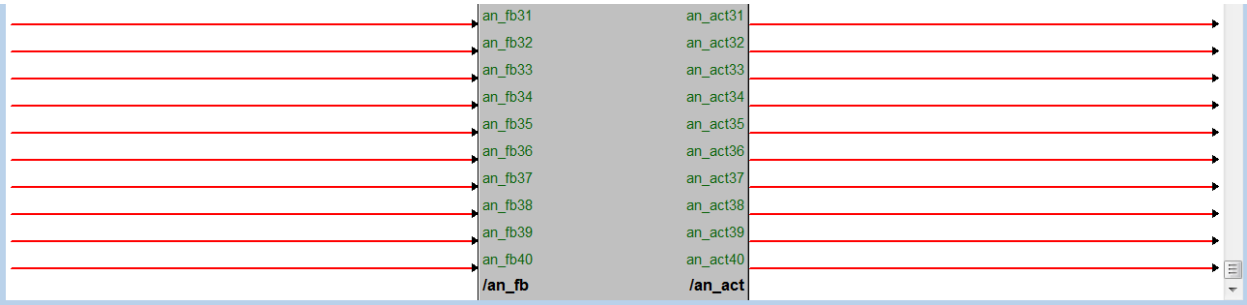
Keeping the Number of Subpage References = 5 and then compiling the VTPRO project, creates the following SGD for SIMPL Windows:

Digitals



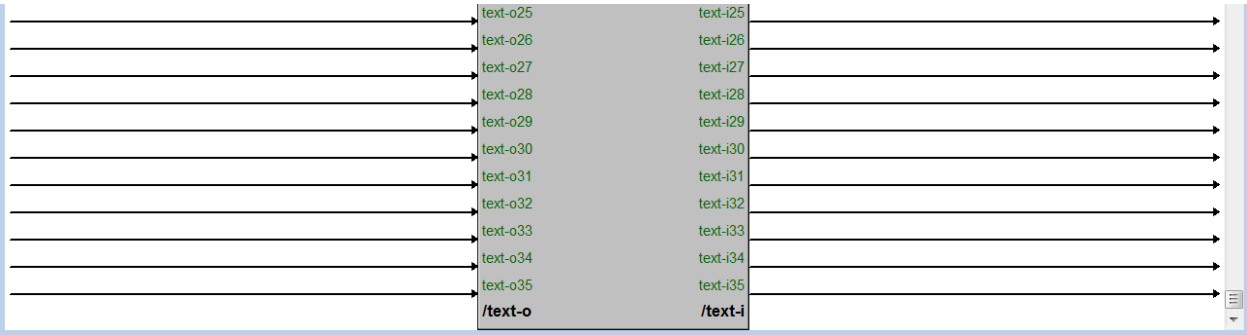
- The digital joins have now increased from 20 to 45.
- press/fb 1 – 9 are associated with the first list item. press/fb 1 – 4 are mapped to Buttons 1 – 4 of this list item.
- press/fb 10 – 18 are associated with the second list item. press/fb 10 – 13 are mapped to Buttons 1 – 4 of this list item.
- Etc.

Analogs



- The analog joins have now increased from 15 to 40.
- an_act/fb 1 – 8 are associated with the first list item. an_act/fb 1 – 3 are mapped to Gauges 1 – 3 of this list item.
- an_act/fb 9 – 16 are associated with the second list item. an_act/fb 9 – 11 are mapped to Gauges 1 – 3 of this list item.
- Etc.

Serials



- The serial joins have now increased from 10 to 35.
- text-i/o 1 – 7 are associated with the first list item. text-i/o 1 – 2 are mapped to Labels 1 – 2 of this list item.
- text-i/o 8 – 14 are associated with the second list item. text-i/o 8 – 9 are mapped to Labels 1 – 2 of this list item.
- Etc.

6 SUPPORTED OBJECTS

Only Non-SmartObjects are supported and only the objects in the following list:

Buttons

- Advanced Button
- Checkbox Button
- Multi-Mode Button
- Simple Button

Sliders

- Advanced Slider Horizontal
- Advanced Slider Vertical
- Fader Slider Horizontal
- Fader Slider Vertical

Images

- Border
- Fill Border
- Image Object

Gauges

- Curtain Controller
- Liquid Gauge Horizontal
- Liquid Gauge Vertical
- Horizontal Segmented Gauge
- Vertical Segmented Gauge
- Shade Controller
- Signed Liquid Gauge Horizontal
- Signed Liquid Gauge Vertical
- WiFi Signal Level Gauge
- Signal Level Gauge

Text

- Formatted Text
- Simple Label
- Text Entry